DATA ANALYST PORTFOLIO PROJECT

ONLINE RETAIL RETURN RISK & PROFITABILITY ANALYSIS

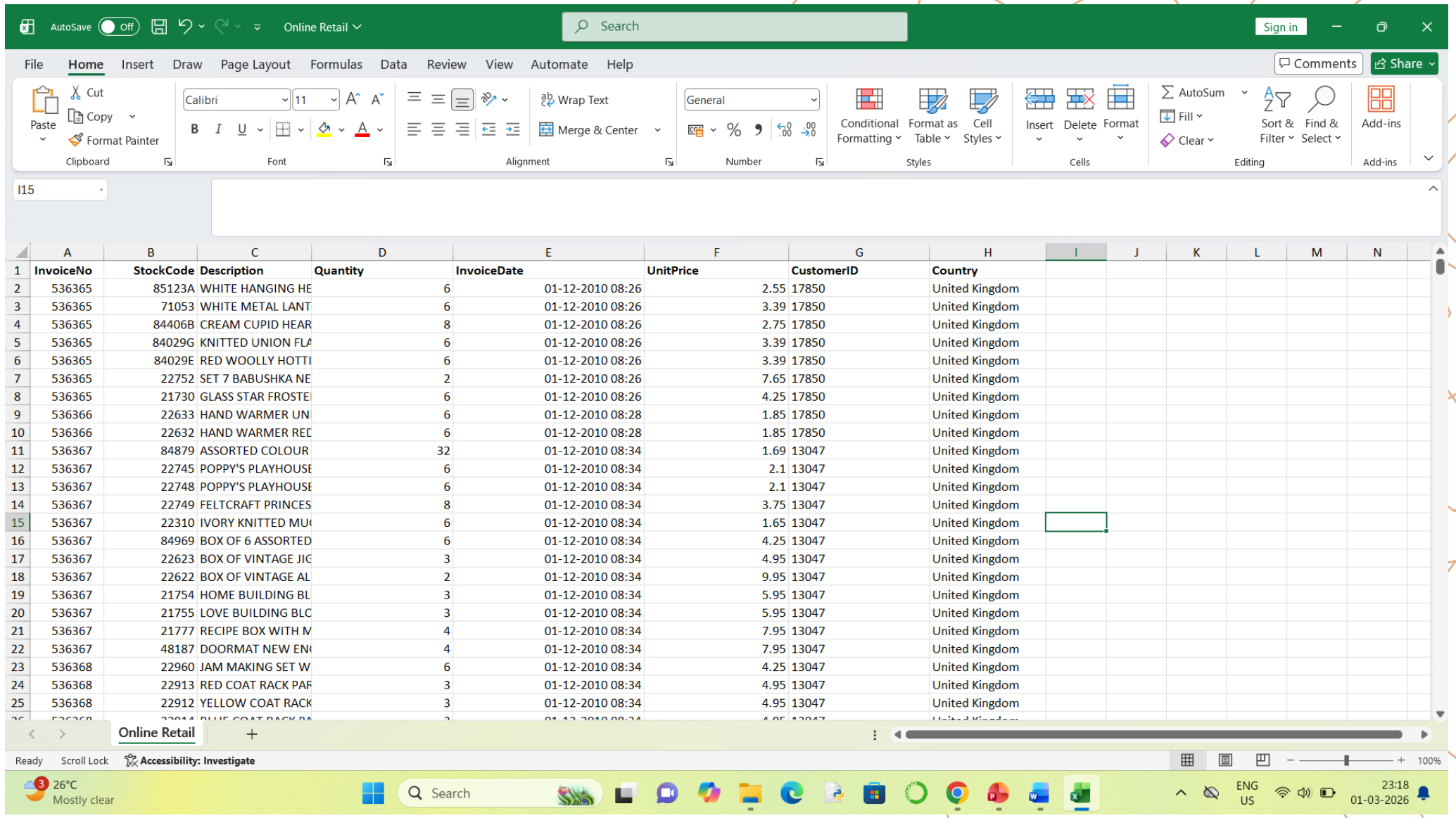
Using Excel, SQL and Power BI

BY

Pratiksha L Rohankar

EXCEL

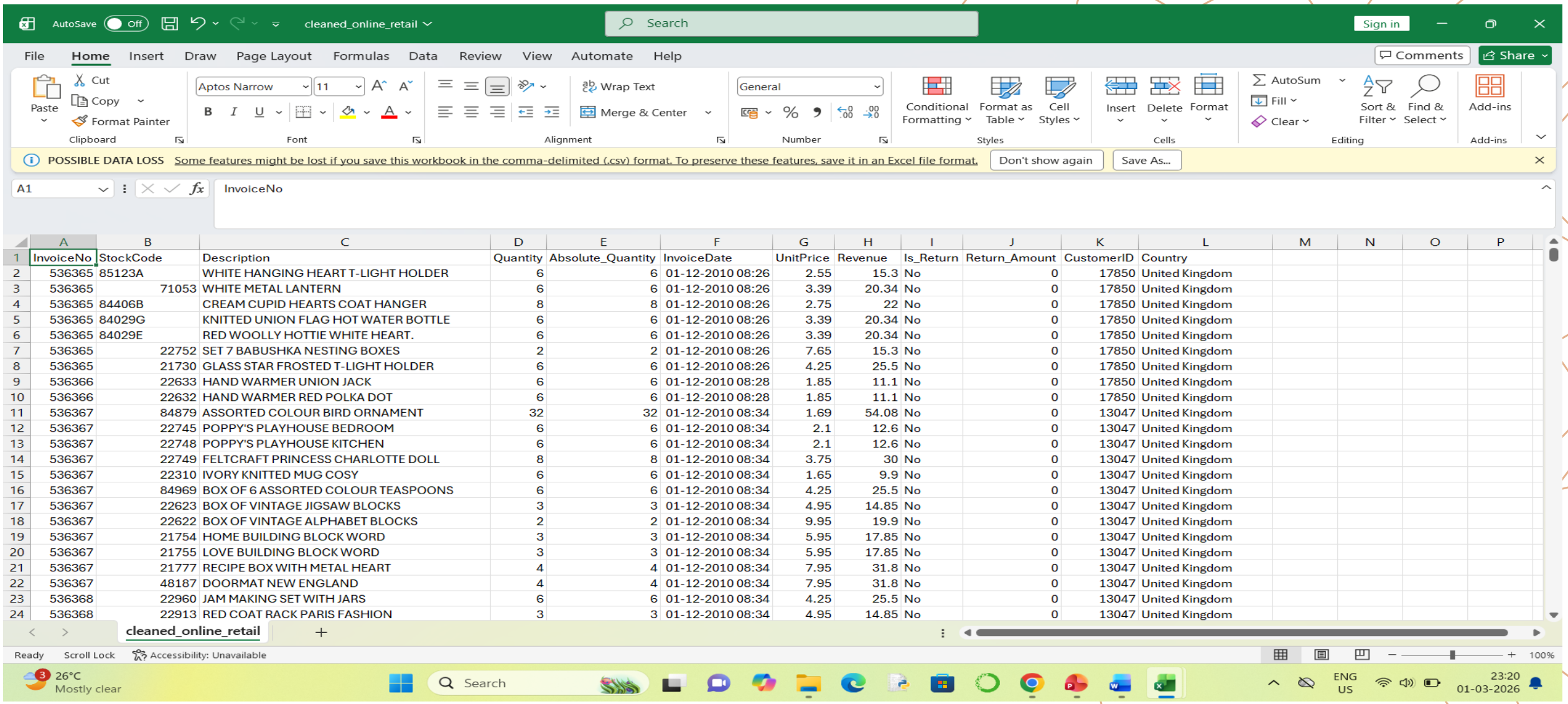
OPEN RAW DATA



InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
536365	85123A	WHITE HANGING HE	6	01-12-2010 08:26	2.55	17850	United Kingdom
536365	71053	WHITE METAL LANT	6	01-12-2010 08:26	3.39	17850	United Kingdom
536365	84406B	CREAM CUPID HEAR	8	01-12-2010 08:26	2.75	17850	United Kingdom
536365	84029G	KNITTED UNION FLA	6	01-12-2010 08:26	3.39	17850	United Kingdom
536365	84029E	RED WOOLLY HOTTI	6	01-12-2010 08:26	3.39	17850	United Kingdom
536365	22752	SET 7 BABUSHKA NE	2	01-12-2010 08:26	7.65	17850	United Kingdom
536365	21730	GLASS STAR FROSTE	6	01-12-2010 08:26	4.25	17850	United Kingdom
536366	22633	HAND WARMER UN	6	01-12-2010 08:28	1.85	17850	United Kingdom
536366	22632	HAND WARMER RED	6	01-12-2010 08:28	1.85	17850	United Kingdom
536367	84879	ASSORTED COLOUR	32	01-12-2010 08:34	1.69	13047	United Kingdom
536367	22745	POPPY'S PLAYHOUSE	6	01-12-2010 08:34	2.1	13047	United Kingdom
536367	22748	POPPY'S PLAYHOUSE	6	01-12-2010 08:34	2.1	13047	United Kingdom
536367	22749	FELTCRAFT PRINCES	8	01-12-2010 08:34	3.75	13047	United Kingdom
536367	22310	IVORY KNITTED MU	6	01-12-2010 08:34	1.65	13047	United Kingdom
536367	84969	BOX OF 6 ASSORTED	6	01-12-2010 08:34	4.25	13047	United Kingdom
536367	22623	BOX OF VINTAGE JIG	3	01-12-2010 08:34	4.95	13047	United Kingdom
536367	22622	BOX OF VINTAGE AL	2	01-12-2010 08:34	9.95	13047	United Kingdom
536367	21754	HOME BUILDING BL	3	01-12-2010 08:34	5.95	13047	United Kingdom
536367	21755	LOVE BUILDING BLC	3	01-12-2010 08:34	5.95	13047	United Kingdom
536367	21777	RECIPE BOX WITH M	4	01-12-2010 08:34	7.95	13047	United Kingdom
536367	48187	DOORMAT NEW EN	4	01-12-2010 08:34	7.95	13047	United Kingdom
536368	22960	JAM MAKING SET W	6	01-12-2010 08:34	4.25	13047	United Kingdom
536368	22913	RED COAT RACK PAR	3	01-12-2010 08:34	4.95	13047	United Kingdom
536368	22912	YELLOW COAT RACK	3	01-12-2010 08:34	4.95	13047	United Kingdom

EXCEL

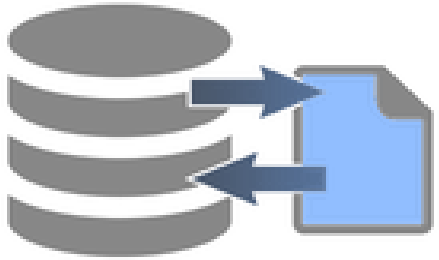
DATA PREPARATION

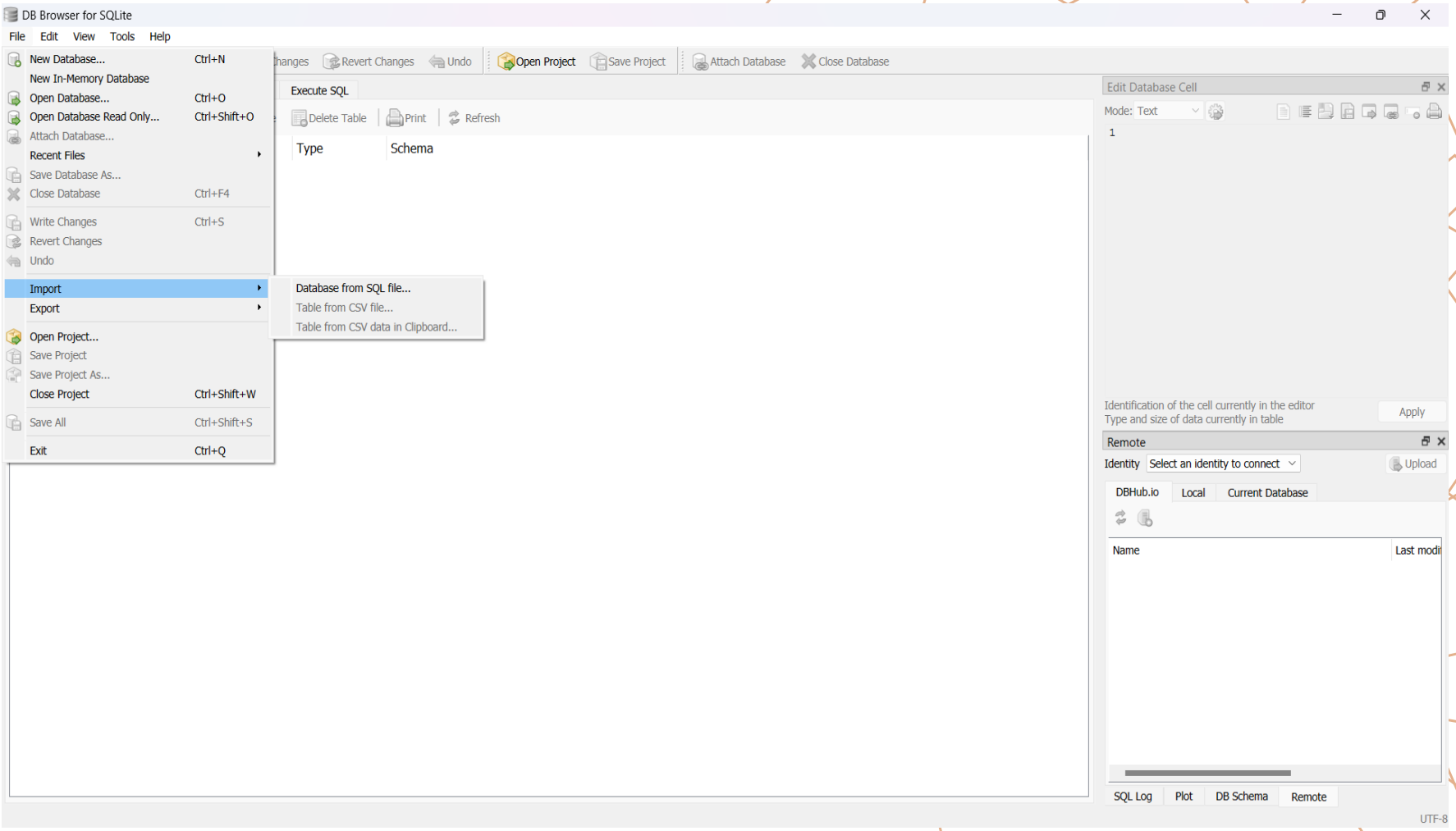


	A	B	C	D	E	F	G	H	I	J	K	L
1	InvoiceNo	StockCode	Description	Quantity	Absolute_Quantity	InvoiceDate	UnitPrice	Revenue	Is_Return	Return_Amount	CustomerID	Country
2	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	6	01-12-2010 08:26	2.55	15.3	No	0	17850	United Kingdom
3	536365	71053	WHITE METAL LANTERN	6	6	01-12-2010 08:26	3.39	20.34	No	0	17850	United Kingdom
4	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	8	01-12-2010 08:26	2.75	22	No	0	17850	United Kingdom
5	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	6	01-12-2010 08:26	3.39	20.34	No	0	17850	United Kingdom
6	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	6	01-12-2010 08:26	3.39	20.34	No	0	17850	United Kingdom
7	536365	22752	SET 7 BABUSHKA NESTING BOXES	2	2	01-12-2010 08:26	7.65	15.3	No	0	17850	United Kingdom
8	536365	21730	GLASS STAR FROSTED T-LIGHT HOLDER	6	6	01-12-2010 08:26	4.25	25.5	No	0	17850	United Kingdom
9	536366	22633	HAND WARMER UNION JACK	6	6	01-12-2010 08:28	1.85	11.1	No	0	17850	United Kingdom
10	536366	22632	HAND WARMER RED POLKA DOT	6	6	01-12-2010 08:28	1.85	11.1	No	0	17850	United Kingdom
11	536367	84879	ASSORTED COLOUR BIRD ORNAMENT	32	32	01-12-2010 08:34	1.69	54.08	No	0	13047	United Kingdom
12	536367	22745	POPPY'S PLAYHOUSE BEDROOM	6	6	01-12-2010 08:34	2.1	12.6	No	0	13047	United Kingdom
13	536367	22748	POPPY'S PLAYHOUSE KITCHEN	6	6	01-12-2010 08:34	2.1	12.6	No	0	13047	United Kingdom
14	536367	22749	FELTCRAFT PRINCESS CHARLOTTE DOLL	8	8	01-12-2010 08:34	3.75	30	No	0	13047	United Kingdom
15	536367	22310	IVORY KNITTED MUG COSY	6	6	01-12-2010 08:34	1.65	9.9	No	0	13047	United Kingdom
16	536367	84969	BOX OF 6 ASSORTED COLOUR TEASPOONS	6	6	01-12-2010 08:34	4.25	25.5	No	0	13047	United Kingdom
17	536367	22623	BOX OF VINTAGE JIGSAW BLOCKS	3	3	01-12-2010 08:34	4.95	14.85	No	0	13047	United Kingdom
18	536367	22622	BOX OF VINTAGE ALPHABET BLOCKS	2	2	01-12-2010 08:34	9.95	19.9	No	0	13047	United Kingdom
19	536367	21754	HOME BUILDING BLOCK WORD	3	3	01-12-2010 08:34	5.95	17.85	No	0	13047	United Kingdom
20	536367	21755	LOVE BUILDING BLOCK WORD	3	3	01-12-2010 08:34	5.95	17.85	No	0	13047	United Kingdom
21	536367	21777	RECIPE BOX WITH METAL HEART	4	4	01-12-2010 08:34	7.95	31.8	No	0	13047	United Kingdom
22	536367	48187	DOORMAT NEW ENGLAND	4	4	01-12-2010 08:34	7.95	31.8	No	0	13047	United Kingdom
23	536368	22960	JAM MAKING SET WITH JARS	6	6	01-12-2010 08:34	4.25	25.5	No	0	13047	United Kingdom
24	536368	22913	RED COAT RACK PARIS FASHION	3	3	01-12-2010 08:34	4.95	14.85	No	0	13047	United Kingdom

DB Browser for SQLite

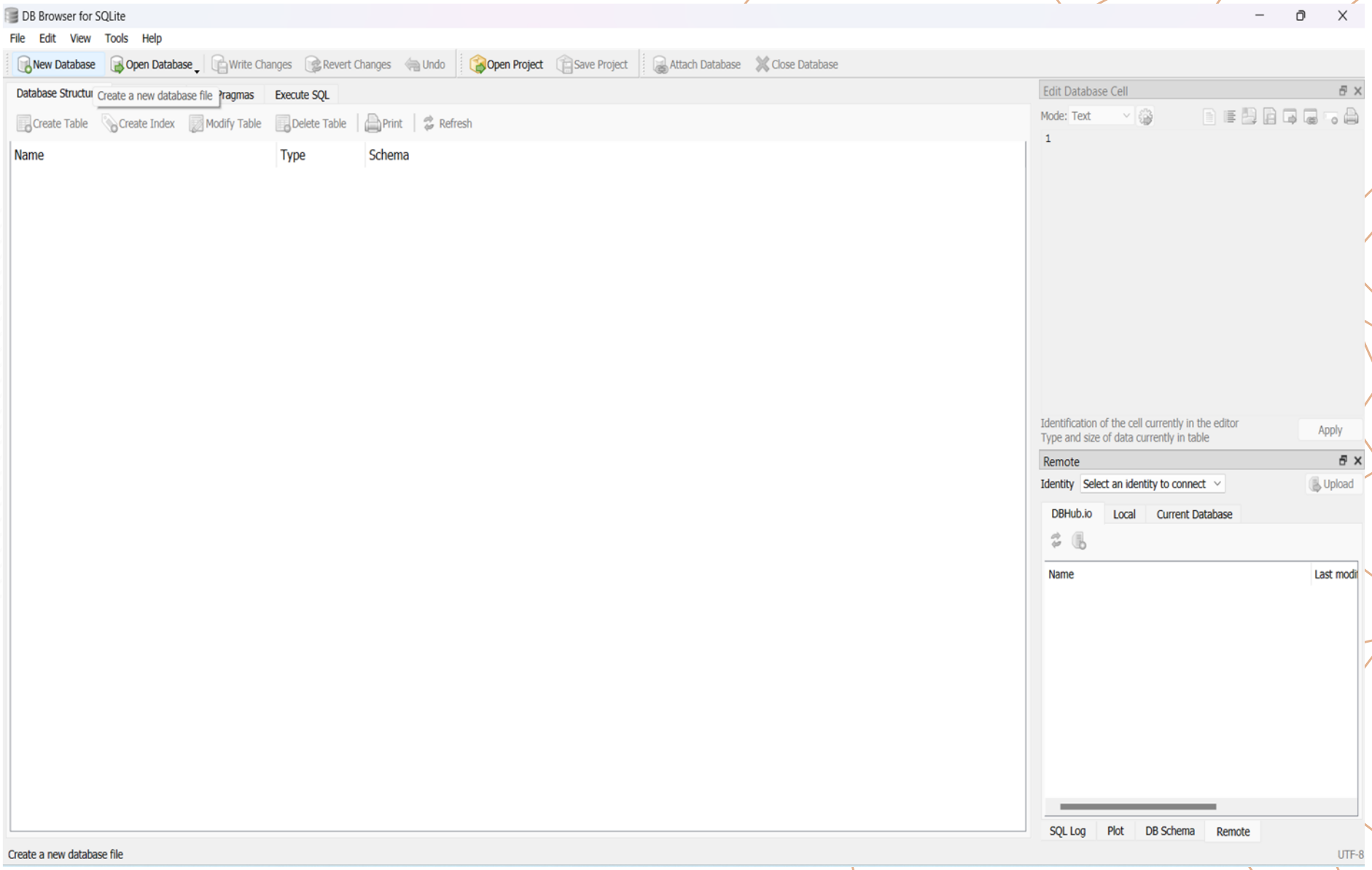
IMPORT DATA





DB Browser for SQLite

CREATING DB



DB Browser for SQLite

WRITING QUERIE

DB Browser for SQLite - C:\Users\rohan\Downloads\My Data Analysis Projects\Practice Project\online_retail.sqbpro [C:\Users\rohan\Downloads\My Data Analysis Projects\online_retail.db]

File Edit View Tools Help

New Database | Open Database | Write Changes | Revert Changes | Undo | Open Project | Save Project | Attach Database | Close Database

Database Structure | Browse Data | Edit Pragmas | Execute SQL

SQL 1*

```sql
SELECT * FROM online_retail;

--Total Revenue
SELECT ROUND(SUM(Revenue), 2) AS total_sales
FROM online_retail
WHERE Quantity > 0;

--Total Return Loss
SELECT
ROUND(SUM(Return_Amount), 2) AS total_return_loss
FROM online_retail;

--Net Revenue After Returns
SELECT
ROUND(SUM(Revenue) - SUM(Return_Amount), 2) AS net_revenue
FROM online_retail
WHERE Quantity > 0;
```

	InvoiceNo	StockCode	Description	Quantity	Absolute_Quantity	InvoiceDate	UnitPrice	Revenue	Is_Return	Return_Amount	CustomerID	Country
1	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	6	01-12-2010 08:26	2.55	15.3	No	0.0	17850	United Ki
2	536365	71053	WHITE METAL LANTERN	6	6	01-12-2010 08:26	3.39	20.34	No	0.0	17850	United Ki
3	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	8	01-12-2010 08:26	2.75	22.0	No	0.0	17850	United Ki
4	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	6	01-12-2010 08:26	3.39	20.34	No	0.0	17850	United Ki
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7	536365	21730	GLASS STAR FROSTED T-LIGHT HOLDER	6	6	01-12-2010 08:26	4.25	25.5	No	0.0	17850	United Ki
8	536366	22633	HAND WARMER UNION JACK	6	6	01-12-2010 08:28	1.85	11.1	No	0.0	17850	United Ki
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12	536367	22748	POPPY'S PLAYHOUSE KITCHEN	6	6	01-12-2010 08:34	2.1	12.6	No	0.0	13047	United Ki
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Edit Database Cell

Mode: Text

1 1

Editing row=1, column=0
Type: Text / Numeric; Size: 1 character(s)

Apply

Remote

Identity: Select an identity to conn

Upload

DBHub.io | Local | Current Database

Name

SQL Log | Plot | DB Schema | Remote

UTF-8

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22:43 01-03-2026

POWER BI

CONNECTING Power BI To DB Browser for SQLite

DATA PROCESSING

DATA VISUALIZATION

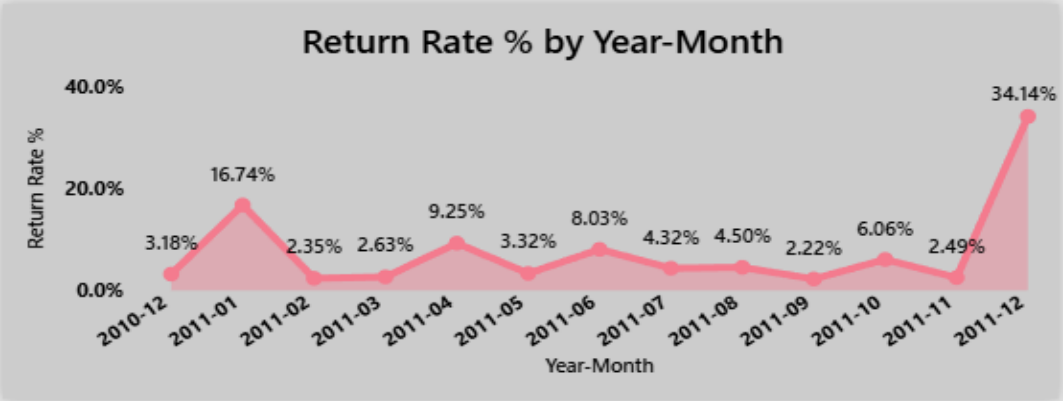
POWER BI

Online Retail Return Risk & Profitability Analysis : OVERVIEW

Total Sales	Net Revenue	Return Rate	Total Return Loss
8.81M	8.21M	6.78%	597.6K

Overview

Risk Drivers



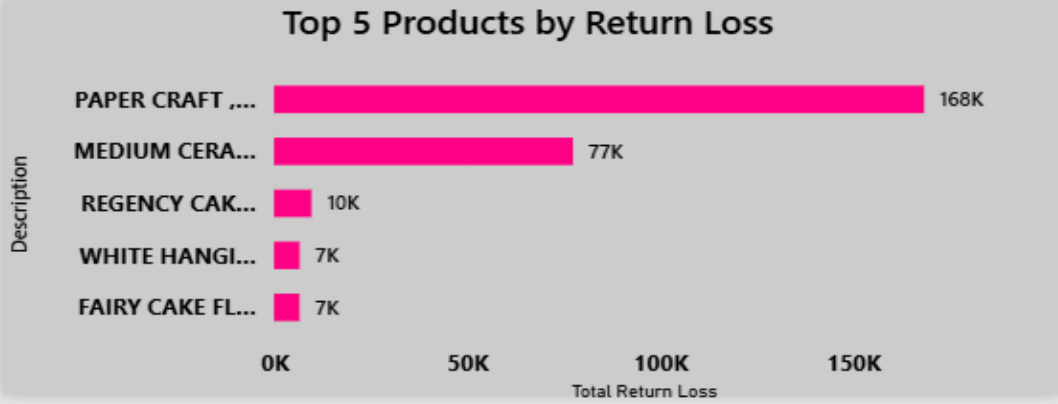
Sales are strong overall, but most return losses come from a few countries and products. The December spike happened mainly because of high returns from the Paper Craft product.

Countries
All

Months
All

Categories
All

Is Return
All



POWER BI

Online Retail Return Risk & Profitability Analysis : RISK DRIVERS

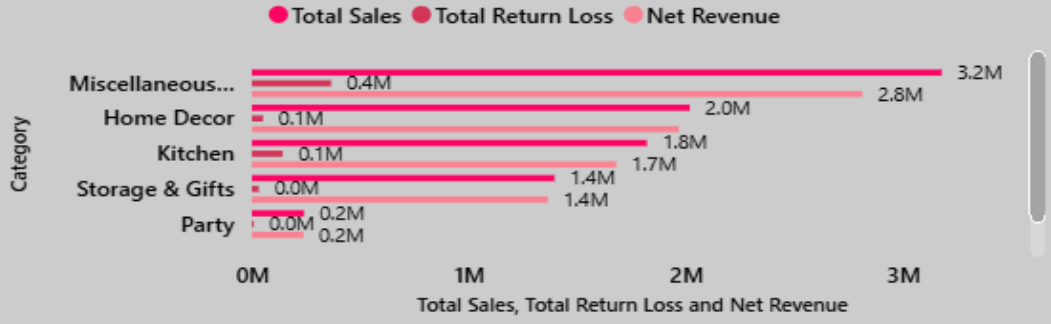
Total Sales	Net Revenue	Return Rate	Total Return Loss
8.81M	8.21M	6.78%	597.6K

Overview

Risk Drivers

Customer Risk Quadrant Analysis

Category Performance Overview



Return losses are mainly driven by a small group of high-risk customers and select product categories, which significantly impact overall profitability.

Countries
All

Months
All

Categories
All

Is Return
All

Return Driver Analysis

Category	Australia	Austria	Belgium	Brazil	Canada	Channel Islands	Cyprus	Denmark	EIRE	European Community	Finland	France	Germany
Home Decor	1.12%		1.84%			1.57%	0.92%	0.27%	4.30%		0.00%	1.06%	1.59%
Kitchen	0.63%	1.22%	0.82%		0.00%	0.17%	1.63%	0.00%	5.23%		1.14%	1.62%	4.10%
Miscellaneous Low Volume Items	1.38%	0.00%	0.51%		0.00%	4.04%	10.23%	1.67%	8.34%		0.01%	15.15%	4.82%
Party	3.20%	0.00%	0.00%					0.77%	0.24%			0.35%	0.77%
Storage & Gifts	0.13%	0.00%	0.55%			0.00%		0.30%	2.46%		0.08%	0.96%	1.79%
Toys	0.05%		0.29%						9.08%			0.41%	0.24%
Total	0.86%	0.51%	0.72%	0.00%	0.00%	1.78%	4.78%	0.93%	5.75%	0.73%	0.22%	6.16%	3.37%

PROBLEM STATEMENT

Despite generating strong total sales of **8.81M**, the company is facing significant profit reduction due to **high product return losses (597.6K)**.

The overall return rate of **6.78%** appears moderate, but deeper analysis shows that return losses are:

- Concentrated in **specific months**, especially a major spike in **December**
- Driven by **a few high-return products**, particularly *Paper Craft*
- Highly impacted by **one major country (United Kingdom)**
- Influenced by a **small group of high-risk customers**
- Associated with certain **product categories**

DASHBOARD 1: SUMMARY

Key Performance Indicators(KPI's) Requirement

Total Sales: Track overall sales revenue with **MTD (Month-To-Date)** and **MoM (Month-over-Month)** growth to monitor business performance trends.

Net Revenue: Measure actual revenue after deducting return losses and analyze monthly profitability trends.

Return Rate: Calculate the percentage of returned items out of total sales and track monthly fluctuations to identify return risk patterns.

Total Return Loss: Monitor total financial loss caused by product returns and analyze month-wise spikes (especially seasonal impacts like December).

Monthly Return Loss Trend: Analyze **Year-Month return loss trends** to detect seasonal spikes and abnormal return behavior.

Monthly Return Rate Trend: Track monthly return rate percentage to identify high-risk periods.

Country-Level Return Loss: Measure total return loss by country to identify **geographical risk concentration**.

Product-Level Return Loss: Track top products contributing to return loss to identify **high-risk items** (e.g., Paper Craft).

DASHBOARD 1: SUMMARY

Key Performance Indicators(KPI's) Requirement

Total Sales: **Track overall sales** to measure revenue exposure across customers and categories.

Net Revenue: **Monitor profit** after returns to evaluate financial impact.

Return Rate %: **Measure return percentage** to identify overall risk levels.

Total Return Loss: **Trak total loss** from returns to understand cost impact.

Customer Risk Segmentation: Classify customers into **High, Moderate, and Low** risk based on return behavior.

Customer Risk Quadrant Analysis: Compare Sales vs Return Loss to identify **high-risk and profitable** customers.

Category Performance Overview: Analyze Sales, Return Loss, and Net Revenue by category.

Return Driver Heatmap: Identify **high-return countries and categories** driving risk

THANK YOU